



**S.B. JAIN INSTITUTE OF TECHNOLOGY
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PostLab

Aim: Study of Fragmentation in Operating System

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❖ **Aim:** Study of Fragmentation in Operating System

❖ **Objectives:**

1. To understand the concept of fragmentation in memory management.
2. To study different types of fragmentation.
3. To analyze internal and external fragmentation.
4. To understand how fragmentation affects memory utilization.
5. To learn techniques used to reduce fragmentation.

❖ **Requirements:**

- Operating System: Windows / Linux
- Tool: Any C/C++ compiler (optional if program used)
- Lab Manual

❖ **Prerequisite:**

Before performing the Fragmentation in Operating System experiment, the following concepts should be known:

Basic understanding of Operating System memory management

Knowledge of contiguous memory allocation

Understanding of process and memory blocks

Familiarity with internal and external fragmentation concept

Basic knowledge of C programming language (variables, loops, arrays)

Understanding of compilation and execution of C programs

Basic idea of memory allocation strategies (First Fit, Best Fit, Worst Fit)

❖ **Theory:**

Fragmentation occurs when memory is divided into small pieces and becomes inefficient to use. It happens when memory is allocated and deallocated dynamically.

Due to fragmentation, available memory exists but cannot be fully utilized. This reduces system performance and increases memory wastage.

Fragmentation mainly occurs in contiguous memory allocation methods.

There are two types:

Internal Fragmentation

External Fragmentation

Types of Fragmentation

1. Internal Fragmentation

Occurs when memory allocated is greater than required memory.

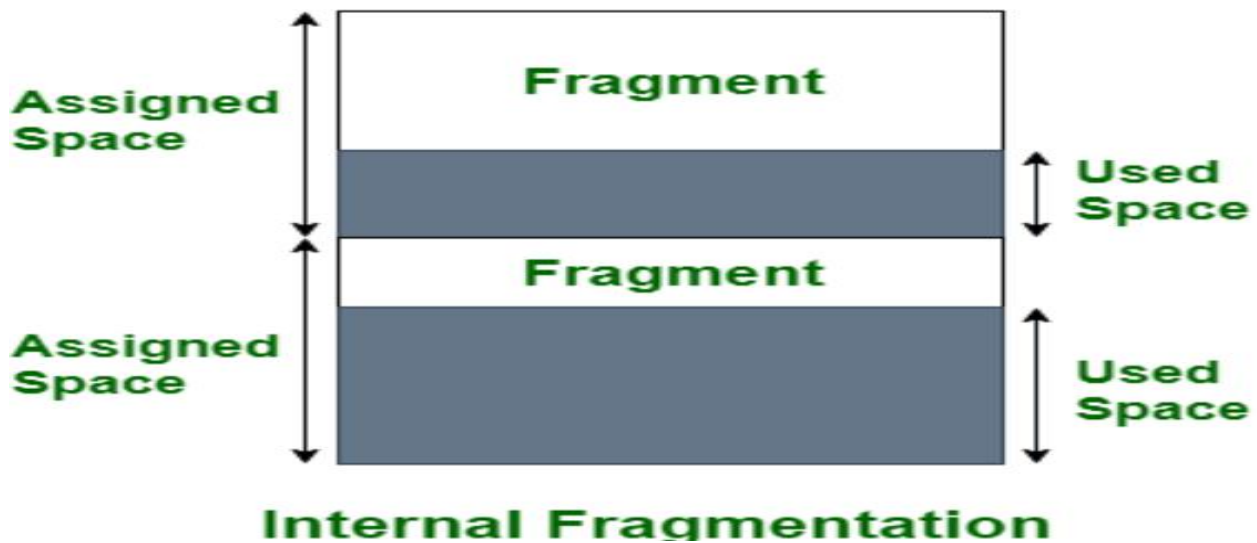
Extra unused memory inside allocated block is wasted.

Happens in fixed partition allocation.

Memory waste occurs within allocated space.

Example:

Process needs 18KB but allocated block = 20KB → 2KB wasted.



2. External Fragmentation

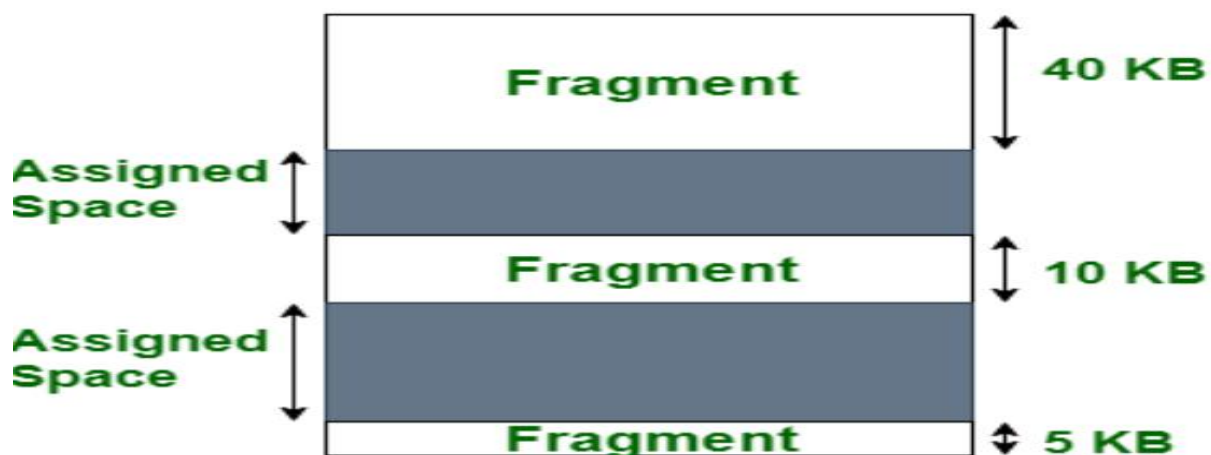
Occurs when free memory is available but not contiguous.

Total memory is sufficient but cannot allocate due to scattered holes.

Happens in variable partition allocation.

Example:

Free blocks: 10KB + 5KB + 8KB → Process needs 20KB → cannot allocate.



Comparison Table

Feature	Internal Fragmentation	External Fragmentation
Memory waste	Inside allocated block	Between memory blocks
Occurs in	Fixed partition	Variable partition
Solution	Better block size	Compaction
Memory usage	Less efficient	Very inefficient
Example	Block bigger than process	Scattered free memory

❖ **Code :**

```
#include<stdio.h>

int main()
{
    int blockSize[10], processSize[10];
    int m, n, i, j;
    int allocation[10], internalFrag = 0;
    printf("Enter number of memory blocks: ");
    scanf("%d", &m);

    printf("Enter size of each block:\n");
    for(i = 0; i < m; i++)
    {
        printf("Block %d: ", i+1);
        scanf("%d", &blockSize[i]);
    }

    printf("\nEnter number of processes: ");
    scanf("%d", &n);
    printf("Enter size of each process:\n");
```

```
for(i = 0; i < n; i++)
{
    printf("Process %d: ", i+1);
    scanf("%d", &processSize[i]);
}

for(i = 0; i < n; i++)
{
    allocation[i] = -1;
}
for(i = 0; i < n; i++)
{
    for(j = 0; j < m; j++)
    {
        if(blockSize[j] >= processSize[i])
        {
            allocation[i] = j;
            internalFrag += blockSize[j] - processSize[i];
            blockSize[j] = 0;
            break;
        }
    }
}

printf("\nProcess No.\tProcess Size\tBlock No.\n");
for(i = 0; i < n; i++)
{
    printf("%d\t%d\t", i+1, processSize[i]);
    if(allocation[i] != -1)
        printf("%d\n", allocation[i]+1);
    else
        printf("Not Allocated\n");
}

printf("\nTotal Internal Fragmentation = %d", internalFrag);
return 0;
}
```

❖ Output:

Output

Enter number of memory blocks: 3

Enter size of each block:

Block 1: 100

Block 2: 500

Block 3: 200

Enter number of processes: 3

Enter size of each process:

Process 1: 90

Process 2: 417

Process 3: 165

Process No.	Process Size	Block No.
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1	90	1
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2	417	2
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3	165	3
---	-----	---

Total Internal Fragmentation = 128

=== Code Execution Successful ===

❖ **Conclusion:**

Fragmentation reduces memory utilization. Internal fragmentation wastes memory inside blocks, while external fragmentation wastes memory between blocks. Proper memory allocation techniques can reduce fragmentation.

❖ **Discussion Questions:**

1. What is fragmentation in operating system?
2. What is internal fragmentation?
3. What is external fragmentation?
4. How can external fragmentation be reduced?
5. What is the main difference between internal and external fragmentation?

❖ **References:**

<https://www.geeksforgeeks.org/operating-systems/what-is-fragmentation-in-operating-system/>

Date: _____ / _____ / **2026**

Signature

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